Task 1 Documentation

IMAGE EDITING TOOL

Using Python, OpenCV & NumPy


Resized


Rotated


Cropped


Color Changed

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Practical Assessment Documentation

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Module Name: Computer Vision

Question: Create a simple image editing tool using Python, OpenCV, and NumPy.

Aim

To design and implement a basic image editing tool that can perform operations such as opening, displaying, saving, resizing, rotating, flipping, cropping, and color manipulation using Python, OpenCV, and NumPy.

Requirements

- Computer System
- Python Software
- Jupyter Notebook / VS Code
- Libraries: OpenCV, NumPy

Steps

1. Import Libraries

Import required libraries for image processing.

2. Load and Display Image

Read the image using and display it using .

3. Save Image

Save the image using .

4. Resize, Rotate, Flip, Crop

Perform basic editing operations.

5. Work with Image Arrays

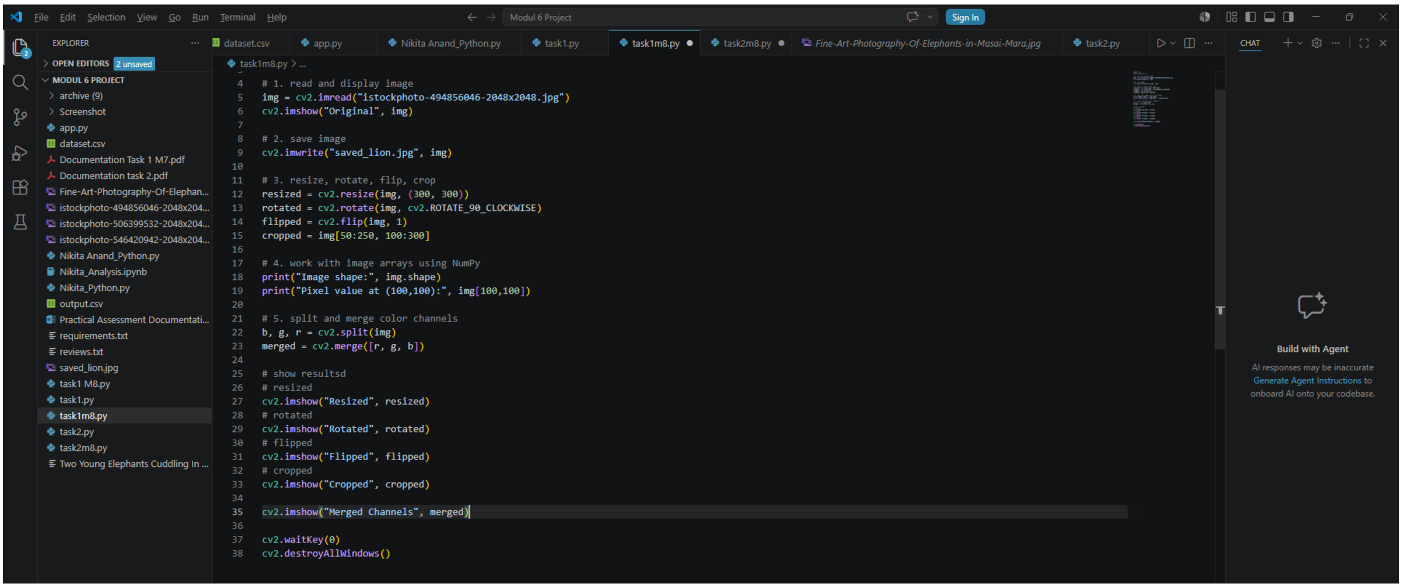
Use NumPy to check image shape and pixel values.

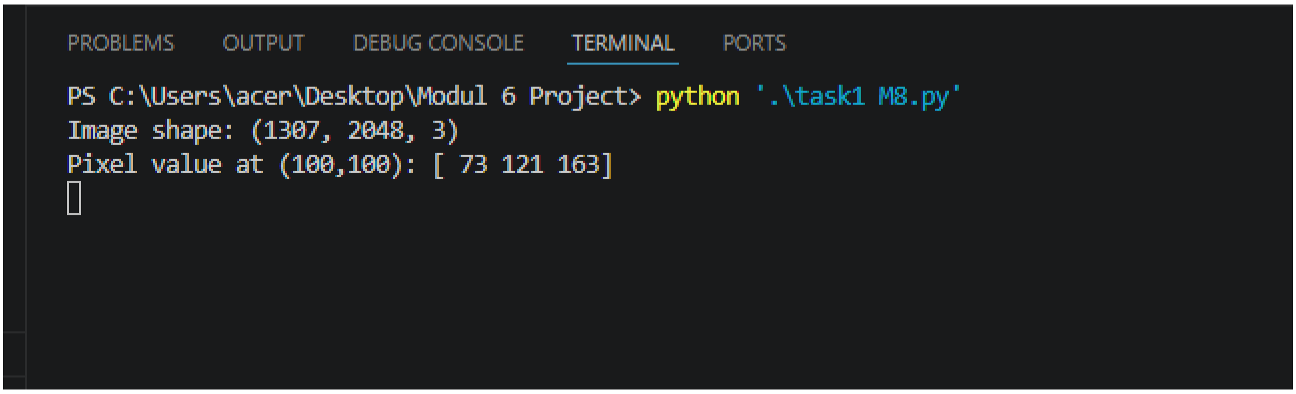
6. Splt and Merge Color Channels

Separate and recombine color channels.

7. Display All Results

Show all edited images in separate windows.

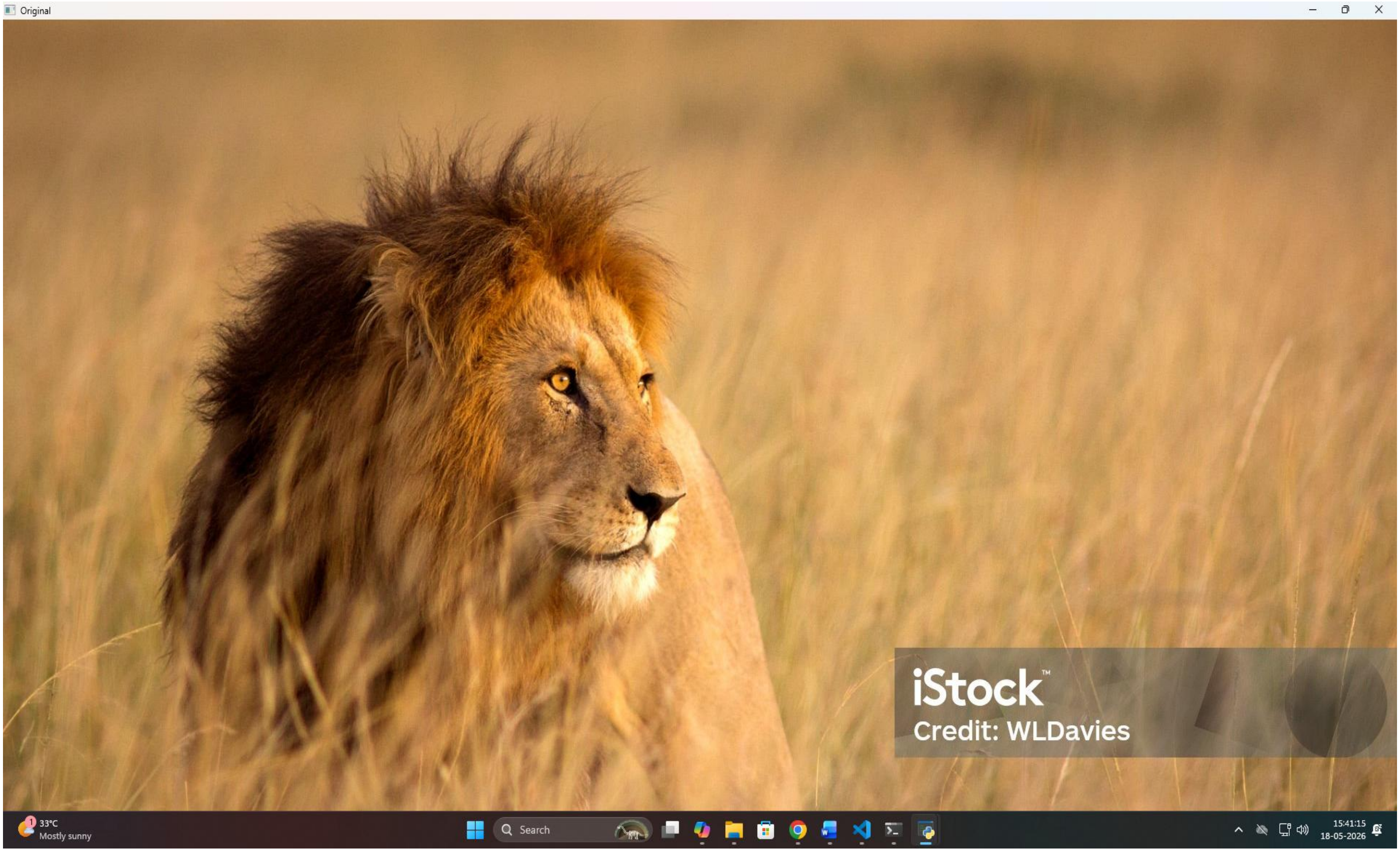




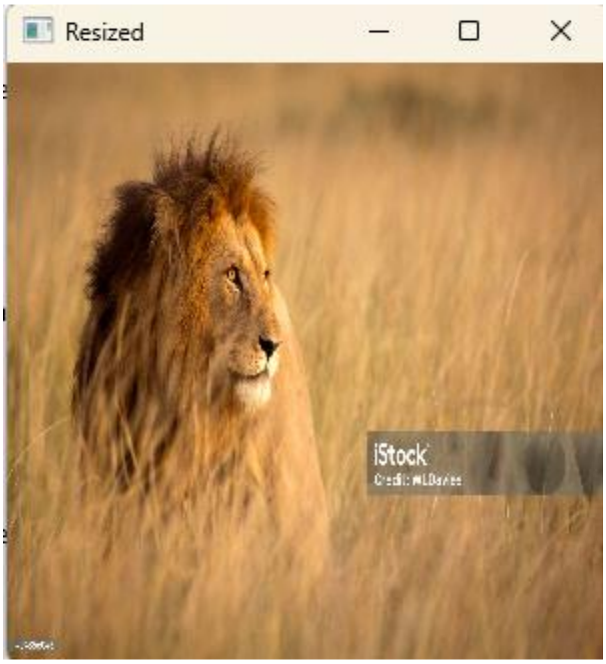
Screenshots

1. Original Image

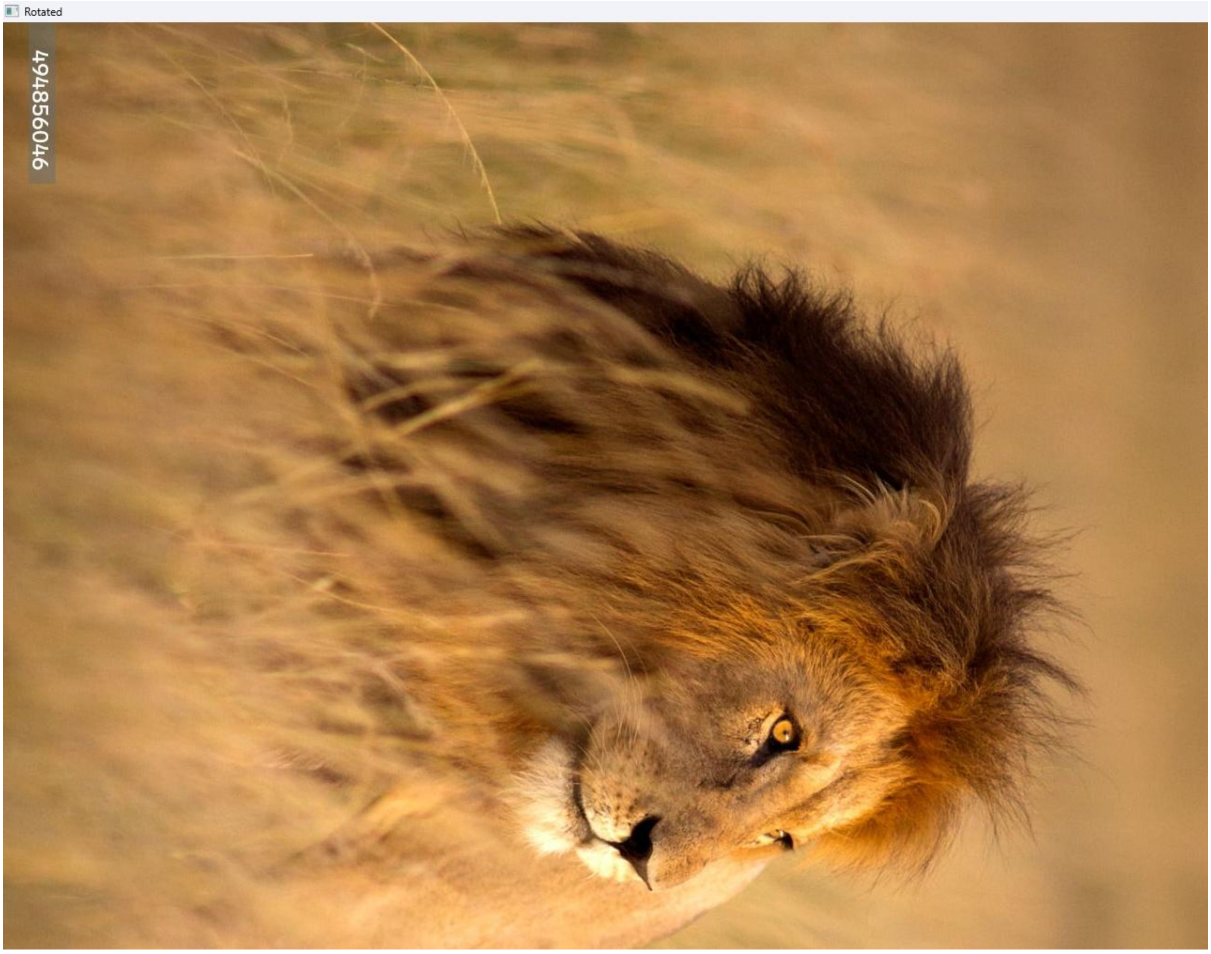
2.



2. Resized Image



3. Rotated Image



4. Flipped Image

5. **Cropped Image**

6. **Color Changed Image**

Results

- Bag of Operations: Successfully performed image editing tasks.
- Resize: Image dimensions changed.
- Rotate: Image rotated 90° clockwise.
- Flip: Image mirrored horizontally.
- Crop: Selected region extracted.
- Color Manipulation: RGB channels swapped successfully.

Conclusion

The project successfully demonstrates how Python, OpenCV, and NumPy can be used to create a simple image editing tool.

It enables basic photo manipulation such as resizing, rotating, flipping, cropping, and color adjustment.

This lays the foundation for developing advanced photo editing applications and understanding image processing concepts.